

RBG-2025-01008

10 CFR 50.73

December 8, 2025

ATTN: Document Control Desk  
U.S. Nuclear Regulatory Commission  
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2025-007-00, Division 1 Balance of Plant  
Isolation Results in Manual Reactor Trip

River Bend Station – Unit 1  
NRC Docket Nos. 50-458  
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report.

This document contains no commitments.

Should you have any questions, please contact Mr. Jack McCoy, Regulatory and Emergency  
Planning Manager, at 225-378-3310.

Respectfully,

**Jack  
McCoy**

Digitally signed by Jack McCoy  
DN: cn=Jack McCoy, o=Manager  
Regulatory and Emergency Preparedness,  
ou=Entergy/ River Bend ,  
email=jmccoy1@entergy.com  
Date: 2025.12.08 08:16:01 -06'00'

JM/jdb

Enclosure: LER 50-458 / 2025-007-00, Division 1 Balance of Plant Isolation  
Results in Manual Reactor Trip

cc: NRC Region IV Regional Administrator - Region IV  
NRC Senior Resident Inspector - River Bend Station

**Enclosure**

**RBG-2025-01008**

Licensee Event Report  
50-458/2025-007-00



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollects.Resource@nrc.gov](mailto:Infocollects.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

|                                                |                                         |                           |                   |
|------------------------------------------------|-----------------------------------------|---------------------------|-------------------|
| 1. Facility Name<br>River Bend Station, Unit 1 | <input checked="" type="checkbox"/> 050 | 2. Docket Number<br>00458 | 3. Page<br>1 of 3 |
|                                                | <input type="checkbox"/> 052            |                           |                   |

4. Title  
Division 1 Balance of Plant Isolation Results in Manual Reactor Trip

| 5. Event Date |     |      | 6. LER Number |                   |         | 7. Report Date |     |      | 8. Other Facilities Involved |                              |               |
|---------------|-----|------|---------------|-------------------|---------|----------------|-----|------|------------------------------|------------------------------|---------------|
| Month         | Day | Year | Year          | Sequential Number | Rev No. | Month          | Day | Year | Facility Name                | <input type="checkbox"/> 050 | Docket Number |
| 10            | 09  | 2025 | 2025          | 007               | 00      | 12             | 08  | 2025 | N/A                          | <input type="checkbox"/> 050 |               |
|               |     |      |               |                   |         |                |     |      | N/A                          | <input type="checkbox"/> 052 |               |

|                        |                        |
|------------------------|------------------------|
| 9. Operating Mode<br>1 | 10. Power Level<br>100 |
|------------------------|------------------------|

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

|                                                    |                                                    |                                                    |                                                        |                                                    |                                     |
|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|-------------------------------------|
| <input checked="" type="checkbox"/> 10 CFR Part 20 | <input type="checkbox"/> 20.2203(a)(2)(vi)         | <input checked="" type="checkbox"/> 10 CFR Part 50 | <input type="checkbox"/> 50.73(a)(2)(ii)(A)            | <input type="checkbox"/> 50.73(a)(2)(viii)(A)      | <input type="checkbox"/> 73.1200(a) |
| <input type="checkbox"/> 20.2201(b)                | <input type="checkbox"/> 20.2203(a)(3)(i)          | <input type="checkbox"/> 50.36(c)(1)(i)(A)         | <input type="checkbox"/> 50.73(a)(2)(ii)(B)            | <input type="checkbox"/> 50.73(a)(2)(viii)(B)      | <input type="checkbox"/> 73.1200(b) |
| <input type="checkbox"/> 20.2201(d)                | <input type="checkbox"/> 20.2203(a)(3)(ii)         | <input type="checkbox"/> 50.36(c)(1)(ii)(A)        | <input type="checkbox"/> 50.73(a)(2)(iii)              | <input type="checkbox"/> 50.73(a)(2)(ix)(A)        | <input type="checkbox"/> 73.1200(c) |
| <input type="checkbox"/> 20.2203(a)(1)             | <input type="checkbox"/> 20.2203(a)(4)             | <input type="checkbox"/> 50.36(c)(2)               | <input checked="" type="checkbox"/> 50.73(a)(2)(iv)(A) | <input type="checkbox"/> 50.73(a)(2)(x)            | <input type="checkbox"/> 73.1200(d) |
| <input type="checkbox"/> 20.2203(a)(2)(i)          | <input checked="" type="checkbox"/> 10 CFR Part 21 | <input type="checkbox"/> 50.46(a)(3)(ii)           | <input type="checkbox"/> 50.73(a)(2)(v)(A)             | <input checked="" type="checkbox"/> 10 CFR Part 73 | <input type="checkbox"/> 73.1200(e) |
| <input type="checkbox"/> 20.2203(a)(2)(ii)         | <input type="checkbox"/> 21.2(c)                   | <input type="checkbox"/> 50.69(g)                  | <input type="checkbox"/> 50.73(a)(2)(v)(B)             | <input type="checkbox"/> 73.77(a)(1)(i)            | <input type="checkbox"/> 73.1200(f) |
| <input type="checkbox"/> 20.2203(a)(2)(iii)        |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(A)         | <input type="checkbox"/> 50.73(a)(2)(v)(C)             | <input type="checkbox"/> 73.77(a)(2)(i)            | <input type="checkbox"/> 73.1200(g) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)         |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(B)         | <input type="checkbox"/> 50.73(a)(2)(v)(D)             | <input type="checkbox"/> 73.77(a)(2)(ii)           | <input type="checkbox"/> 73.1200(h) |
| <input type="checkbox"/> 20.2203(a)(2)(v)          |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(C)         | <input type="checkbox"/> 50.73(a)(2)(vii)              |                                                    |                                     |

☐ Other (Specify here, in abstract, or in NRC 366A).

## 12. Licensee Contact for this LER

|                                                                           |                                                    |
|---------------------------------------------------------------------------|----------------------------------------------------|
| Licensee Contact<br>Jack McCoy, Regulatory and Emergency Planning Manager | Phone Number (Include Area Code)<br>(225) 378-3310 |
|---------------------------------------------------------------------------|----------------------------------------------------|

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable To IRIS | Cause | System | Component | Manufacturer | Reportable To IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| A     |        |           |              | Y                  |       |        |           |              |                    |

## 14. Supplemental Report Expected

|                                        |                                                                              |                              |       |     |      |
|----------------------------------------|------------------------------------------------------------------------------|------------------------------|-------|-----|------|
| <input checked="" type="checkbox"/> No | <input type="checkbox"/> Yes (If yes, complete 15. Expected Submission Date) | 15. Expected Submission Date | Month | Day | Year |
|                                        |                                                                              |                              |       |     |      |

## 16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On October 9, 2025, at 1207 CDT, River Bend Station (RBS) was operating at 100% reactor power when a manual reactor trip was inserted as a result of a Division I Balance of Plant (BOP) isolation. The BOP isolation resulted in the loss of the Instrument Air System (IAS) and the Component Cooling Primary System (CCP) to containment. All plant systems responded as designed.

Immediately following the manual trip, a reactor water low level 3 isolation signal was received as expected. Due to the swell from the main turbine trip, a reactor water high level 8 actuation signal was received at 1208 CDT. The level 8 actuation signal was reset at 1211 CDT.

At 1237 CDT, Control Room Operators entered the Emergency Operating Procedure (EOP) for high containment pressure caused by the isolation of the containment vent flow path. The isolation signal for IAS and CCP was reset at 1255 CDT. Following the restoration of the isolation signal, Control Room Operators lowered containment pressure and exited the EOP at 1305 CDT.

The direct cause of this event was a failed fuse caused by a human performance error.

This report is made pursuant to 10 CFR 50.73(a)(2)(iv)(A), any event or condition that resulted in manual or automatic actuation of the Reactor Protection System.



**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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| 1. FACILITY NAME           |                                         | 2. DOCKET NUMBER | 3. LER NUMBER |                      |            |
|----------------------------|-----------------------------------------|------------------|---------------|----------------------|------------|
| River Bend Station, Unit 1 | <input checked="" type="checkbox"/> 050 | 00458            | YEAR          | SEQUENTIAL<br>NUMBER | REV<br>NO. |
|                            | <input type="checkbox"/> 052            |                  | 2025          | 007                  | 00         |

**NARRATIVE****Event Description**

On October 9, 2025, at 1207 CDT, with River Bend Station (RBS), Unit 1 operating at 100% reactor power, a manual reactor trip was inserted as a result of a Division 1 Balance of Plant (BOP) isolation. The Division 1 BOP isolation resulted in the loss of the Instrument Air System (IAS) [LD] and the Component Cooling Primary System (CCP) [CC] to containment.

On October 9, 2025, at 1203 CDT, a Division 1 BOP isolation signal was received. Control Room operators recognized the loss of IAS and CCP to containment and entered the appropriate procedures. The Control Room operators were unable to reset the Division 1 BOP isolation signal. At 1207 CDT, the Control Room Operators inserted a manual reactor trip by placing the Mode Switch to Shutdown. All plant systems responded as designed.

Immediately following the manual trip, a low reactor water level isolation signal was received as expected. Due to the swell from the Main Turbine [TA] trip, a high reactor water level actuation signal was received at 1208 CDT. The high reactor water level actuation caused the Reactor Feedwater System (FWS) [SJ] to trip as designed. The high reactor water level actuation signal was reset at 1211 CDT and Reactor Feedwater Pump C (FWS-P1C) was restarted to maintain reactor water level. Reactor pressure was maintained by the Turbine Bypass Valves [JI].

At 1237 CDT, Control Room operators entered the Emergency Operating Procedure (EOP) for high containment pressure. The increase in containment pressure was due to the isolation of the containment vent flow path. At 1255 CDT, the isolation signal for IAS and CCP was reset. Following the restoration of the isolation signal, Control Room operators lowered containment pressure and exited the EOP at 1305 CDT.

This event was reported on October 9, 2025, at 1539 CDT in accordance with 10 CFR 50.72(b)(2)(iv)(B) and 10 CFR 50.72(b)(3)(iv)(A) as an event or condition that results in actuation of the RPS when the reactor is critical and as a specified system actuation due to the expected low reactor water level isolation signal immediately following the reactor scram and a subsequent high reactor water level signal (EN 57975).

**Event Cause**

Prior to the Division 1 BOP isolation, plant personnel were performing surveillance STP-058-4201, Containment and Drywell Manual Isolation Actuation Logic System Functional Test. During the surveillance, a technician lifted a lead and did not appropriately account for the length of wire allowing the wire to retract and contact the metal enclosure. This caused the Division 1 Isolation Logic Supply Fuse (B21H-F28A) [JM] to fail. Due to the failure of B21H-F28A, a Division 1 BOP isolation actuation occurred. The Division 1 BOP isolation resulted in the loss of the IAS and CCP to containment.

**Safety Assessment**

The actual consequence of this event was a Division 1 BOP containment isolation that was unable to be reset and a subsequent reactor scram. There were no other actual consequences to general safety of the public, nuclear safety, industrial safety and radiological safety for this event.

**Corrective Actions****Planned:**

- Complete performance management with personnel involved to address personal accountability.
- Develop and implement a non-training intervention plan to reinforce standards and expectations.



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|                            | <input type="checkbox"/> 052            |                  | 2025          | 007               | 00      |

**Previous Occurrences**

**LER-2025-001-00, Human Performance Error Results in Manual Reactor Trip**

Plant personnel failed to close SWP-V3302, SWP-MOV55A block valve, prior to post maintenance testing. During post maintenance testing, maintenance personnel requested Operations to open SWP-MOV55A, Standby Cooling Tower Division 1 Return Isolation Valve, as the valve opened all normal service water pumps received a low-pressure lockout condition due to an open flow path to the Standby Cooling Tower. This resulted in a loss of Normal Service Water and a subsequent manual reactor trip.

**LER-2023-006-00, Automatic Reactor Scram Due to a Turbine Trip Following a Disconnect Failure at Switchyard**

The cause of the turbine trip signal was due to a Generator Side Disconnect switch failing as a result of not being fully seated in the saddle of the disconnect fingers. After further investigation, it was determined that the disconnects were not properly seated in the saddle of the disconnect fingers due to personnel error.

NOTE: Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].